

[slides]

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Uncertainty – The Cost of Choosing

You have 3 options:

- Agent 1 → 95% chance, leave at 7:00 pm
- Agent 2 → 98% chance, leave at 8:00 pm
- Agent 3 → 100% chance, leave at 10:00 pm

👉 You choose **Agent 2 (98% at 8:00 pm)**

Why?

- Agent 3 is useless (you arrive late even though probability = 1)
- Agent 1 is less reliable than Agent 2
- Agent 2 gives **best balance of high probability + arriving on time**

👉 Key idea: **Best decision ≠ highest probability only, it must satisfy the goal too**

Uncertainty Example (Cavity Diagnosis)

Rules in Knowledge Base (KB):

- Toothache $\Rightarrow$ Cavity
- Toothache $\Rightarrow$ Cavity $\vee$ GumProblem $\vee$ Abscess
- Cavity $\Rightarrow$ Toothache

✗ Problem:

- These rules are **not always true**
- **Toothache doesn't always mean cavity**

👉 So logic alone fails because real-world knowledge is **uncertain + incomplete**

Degree of Belief

Instead of true/false, we use probability

Probabilities are a way to summarize uncertainty

Example:

- “There’s 80% chance of cavity”

👉 This means:

- Based on observed facts, cavity is **likely but not certain**
-

Likelihood and Importance

Rational decisions depend on 2 things:

- **Likelihood** → probability of outcome
- **Importance** → utility (how valuable the outcome is)

👉 **Decision Theory = Probability + Utility**

👉 **Maximum Expected Utility (MEU):**

- Choose action with **highest average utility** considering all outcomes
-

! Arguments Against Probabilities in AI

People **criticize probability** because:

- **Needs** lots of **data**
- Must **consider all possibilities**
- **Hides** detailed **reasoning**
- **Humans estimate** probabilities **poorly**
- Exact numbers often unavailable
- Sometimes **inconvenient** to use

👉 But still widely used because it **handles uncertainty better than logic**

📌 Probability

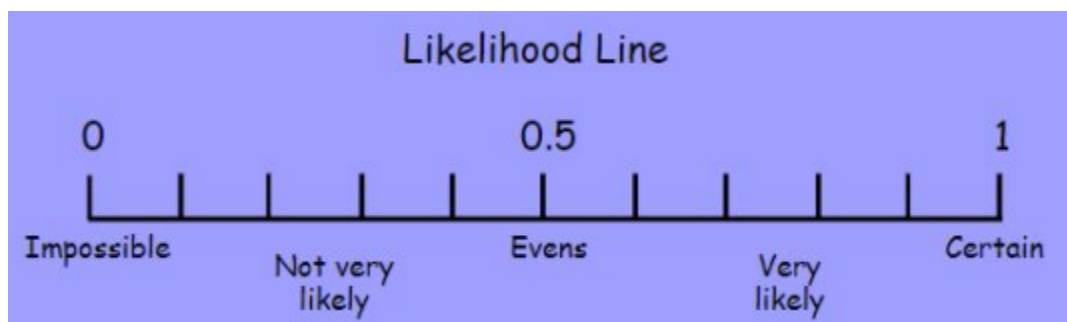
👉 **Probability = measure of how likely an event is**

- Range: **0 to 1**
 - **0** → **impossible**
 - **1** → **certain**
 - **between** → **uncertain**

👉 Example:

- $0.8 = 80\%$ chance

👉 Key idea: **Probability helps AI make decisions when things aren't certain**





Probabilities – Language of Propositions

- **Random Variable** → a **variable** whose **value** is **uncertain**
 - Examples: **die1**, **cavity**, **Weather**
- **Domain** → all **possible values** the **variable** can **take**
 - Example:
 - $\text{die1} = \{1, 2, 3, 4, 5, 6\}$

👉 Domains for given variables:

- Weather = {sunny, rainy, cloudy, snow}
- Cavity = {true, false}
- Age = could be {0,1,2,...} (or ranges like young, adult, old)

👉 Key idea: **Random variable + domain = what values are possible**



Random Variables Explained

- A random variable **can take different values**, each **with a probability**
- Its value changes due to uncertainty

👉 Types:

- **Discrete random variable** → **finite/countable** values
 - Example: dice, weather

👉 Important properties:

- Values must be:
 - **Exhaustive** → cover all possibilities
 - **Mutually exclusive** → only one value at a time

👉 Examples:

- Weather = {sunny, rain, cloudy, snow}
- Boolean variable = {true, false}

👉 Key idea: **Only one value occurs at a time, but all are possible**



Probability Distribution Language

Given:

Weather = {sunny, rainy, cloudy, snow}

Probabilities:

- $P(\text{sunny}) = 0.6$
- $P(\text{rainy}) = 0.1$
- $P(\text{cloudy}) = 0.29$
- $P(\text{snow}) = 0.01$

👉 This is called a **probability distribution**

👉 Ways to write it:

- Table form:

$$P(\text{Weather}) =$$

sunny	rainy	cloudy	snow
0.6	0.1	0.29	0.01

- Vector form:
 - $P(\text{Weather}) = \langle 0.6, 0.1, 0.29, 0.01 \rangle$

👉 Important rule:

- All probabilities must sum to 1

👉 Key idea: **Probability distribution = probability assigned to every possible value**

Probabilities – Concepts

- **Sample Space (Ω)** → **all possible outcomes**
 - Example (dice): $\Omega = \{1, 2, 3, 4, 5, 6\}$
- **ω (omega)** → **one specific outcome from Ω**
 - Example: $\omega = 3$

👉 Rules:

- **$0 \leq P(\omega) \leq 1$** (probability is always between 0 and 1)
- **Sum of all probabilities = 1**
 - **$\sum P(\omega) = 1$**

👉 Basic formula:

- **Probability = (Number of favorable outcomes) / (Total outcomes)**

👉 Examples:

- Coin toss (head):
 - **$P(\text{head}) = 1/2$**
- Odd number on dice:
 - Outcomes = $\{1, 3, 5\}$ → 3 favorable
 - Total = 6
 - **$P(\text{odd}) = 3/6 = 1/2$**
- Fair dice probabilities:
 - **$P(1)=P(2)=P(3)=P(4)=P(5)=P(6)=1/6$**

👉 Key idea: **Probability measures how likely an outcome is within all possibilities**

Prior (Unconditional) Probability

- probability of an event *without any extra information* → just normal prob
- Written as:
 - $P(A)$

Examples:

- **$P(\text{doubles})$** when rolling 2 dice
 - Possible doubles: (1,1), (2,2), ... (6,6) → **6 outcomes**
 - Total outcomes = $6^2 = 36$
 - **$P(\text{doubles}) = 6/36 = 1/6$**

Posterior (Conditional) Probability

- **Updated probability after evidence**
- Written as:
 - **$P(\text{Event} \mid \text{Evidence})$**
 - **$P(A \mid B) \rightarrow$ probability of A given B**

Examples:

- $P(\text{doubles} \mid \text{first die} = 3)$

Step-by-step:

- If **first die = 3**, possible outcomes:
 - (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) → **6 outcomes**
- **Only one is double: (3,3)**

 So:

- **$P(\text{doubles} \mid \text{die1} = 3) = 1/6$**

- $P(\text{cavity} \mid \text{toothache} = \text{true})$

👉 Meaning:

- “Given the person has a toothache, what’s the chance they have a cavity?”

👉 This is **more realistic** than $P(\text{cavity})$ alone



Joint Probability Distribution

- It gives probability for **all combinations of variables**

👉 Table values:

- $P(\text{cavity} \wedge \text{toothache}) = 0.04$
- $P(\text{cavity} \wedge \neg \text{toothache}) = 0.06$
- $P(\neg \text{cavity} \wedge \text{toothache}) = 0.01$
- $P(\neg \text{cavity} \wedge \neg \text{toothache}) = 0.89$

	toothache	$\neg$ toothache
cavity	0.04	0.06
$\neg$ cavity	0.01	0.89

👉 Key idea:

Each cell = probability of BOTH events happening together



Sum Must Be 1

- **All entries** in the table must **add up to 1**

👉 Check:

$$0.04 + 0.06 + 0.01 + 0.89 = 1 \checkmark$$

📌 Answering Any Question Using Joint Distribution

👉 You can compute any probability from this table

- Just **add relevant entries**
-

+ Marginal Probability (Adding Rows/Columns)

- **Probability of a single variable** (ignoring others)

👉 Example 1:

P(cavity)

$$= P(\text{cavity} \wedge \text{toothache}) + P(\text{cavity} \wedge \neg\text{toothache})$$

$$= 0.04 + 0.06 = \mathbf{0.10}$$

👉 Example 2:

P(toothache)

$$= P(\text{cavity} \wedge \text{toothache}) + P(\neg\text{cavity} \wedge \text{toothache})$$

$$= 0.04 + 0.01 = \mathbf{0.05}$$

👉 Key idea:

Marginal = sum over rows or columns

From the Table (Prior vs Conditional)

	toothache	$\neg$ toothache
cavity	0.04	0.06
$\neg$ cavity	0.01	0.89

- $P(\text{cavity}) = 0.1 \rightarrow$ **prior (unconditional)**
- $P(\text{cavity} \wedge \text{toothache}) = 0.04 \rightarrow$ also from joint table

👉 These are **before using evidence**

After Evidence (Posterior / conditional prob)

- Now suppose we know:
 - **toothache = true**

👉 We want:

- $P(\text{cavity} \mid \text{toothache})$

👉 This is **posterior (conditional probability)**

Formula

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

“Probability of a happening given b has already happened”

👉 Apply here:

- $P(\text{cavity} \mid \text{toothache}) = P(\text{cavity} \wedge \text{toothache}) / P(\text{toothache})$
- $= 0.04 / 0.05 = 0.8$

Product Rule (Very Important)

- Rearranging the formula gives:
 - $P(a \wedge b) = P(a | b) \times P(b)$

👉 Also:

$$P(a \wedge b) = P(b | a) \times P(a)$$

👉 Meaning:

- Joint probability can be computed in 2 ways:
 - First b happens, then a
 - First a happens, then b

^ the above 2 formulas are **equivalent!!**

- **Conditional form:**
 - $P(a | b) = P(a \wedge b) / P(b)$
- **Product form:**
 - $P(a \wedge b) = P(a | b) \times P(b)$

👉 They are just **rearrangements of each other**

Generalization to Distributions

- For full variables:
 - $P(\text{Weather}, \text{Cavity}) = P(\text{Weather} | \text{Cavity}) \times P(\text{Cavity})$

👉 Meaning:

- **Joint probability of 2 variables = Conditional probability × prior**

RULE	FORMULA
Product / Multiplication Rule	$P(E \cap F) = P(F) \cdot P(E F)$
General Product Rule	$P(E_1 \cap E_2 \cap \dots \cap E_n) = P(E_1) \cdot P(E_2 E_1) \cdot P(E_3 E_1 \cap E_2) \dots$
Law of Total Probability	$P(E) = \sum_i P(E F_i) \cdot P(F_i)$ for a partition $\{F_1, \dots, F_n\}$

Independent Events – Core Concepts

- **General Rule:**

- $P(A \cap B) = P(A) \times P(B | A)$

👉 Probability of both events = probability of first $\times$ probability of second given first

> Independence Definition:

Events are **independent** if the **occurrence of one does not change the probability of the other**

- $P(A | B) = P(A)$
 - $P(B | A) = P(B)$
-

- **Independent Formula:**

- $P(A \cap B) = P(A) \times P(B)$

👉 Key idea:

If independent $\rightarrow$ intersection is simply the product of their individual probabilities



Practical Example (Bag Problem)

Independent Events - With Replacement.

- $P(A \cap B) = P(A) P(B|A) = P(B) P(A|B)$
- $P(A|B) = P(A) \quad P(B|A) = P(B)$
- $\underline{P(A \cap B) = P(A) P(B)}$ Independent.

2 Red 3 Black $\rightarrow$ One by one (R B)

$$P(RB) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10} \text{ Ans (Without)}$$

$$P(RB) = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25} \text{ (With Replacement)}$$

$$P(RB) + P(BR) = \frac{2}{5} \times \frac{3}{5} + \frac{3}{5} \times \frac{2}{5} = \frac{12}{25} \text{ Ans}$$

- Bag contains:
 - 2 Red (R), 3 Black (B) $\rightarrow$ Total = 5

👉 Find: Probability of picking **Red then Black**

🚫 Scenario A: Without Replacement (**Dependent**)

- First ball is **NOT put back** $\rightarrow$ probabilities change

👉 1st Draw (Red):

- $P(R) = 2/5$

👉 2nd Draw (Black):

- $P(B | R) = 3/4$ (only 4 balls left)

👉 Calculation:

- $P(R \cap B) = (2/5) \times (3/4) = 6/20 = 3/10$

👉 Key idea:

Events are **dependent** because **first draw affects second**

♻️ Scenario B: With Replacement (**Independent**)

- Ball is **put back** → probabilities stay same

👉 1st Draw (Red):

- $P(R) = 2/5$

👉 2nd Draw (Black):

- $P(B) = 3/5$

👉 Calculation:

- $P(R \cap B) = (2/5) \times (3/5) = 6/25$

👉 Key idea:

With replacement → events become **independent**

↔️ Order vs Combination

👉 If question = "one red and one black in ANY order"

You must add both cases:

- Red then Black:

- $P(RB) = 6/25$
- Black then Red:
 - $P(BR) = 6/25$

👉 Total Probability:

- $P = 6/25 + 6/25 = 12/25$

Probabilistic Inference

- It means: **computing probabilities using evidence**
- Full joint distribution = **complete information**
- Inference = **extracting answers from it**

👉 What we do:

- Use **observed evidence**
- Compute **posterior probabilities**
- Answer **queries (questions)**

👉 Example:

- Given toothache, find $P(\text{cavity} \mid \text{toothache})$

👉 Key idea:

Use known data to infer unknown probabilities

Joint Distribution as Knowledge Base

- The **full joint distribution** contains **probabilities** of **all combinations**

👉 It acts like a **knowledge base**

👉 From it, we can compute:

- Marginal probabilities
- Conditional probabilities
- Any query

👉 Key idea:

If you know the full joint distribution → you can answer ANY question

Example Variables

- Toothache (T) → true/false
- Cavity (C) → true/false
- ShowsOnXRay (X) → true/false

Ex: three Boolean variables *Toothache (T), Cavity (C), ShowsOnXRay (X)*

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

👉 Total combinations = $2^3 = 8$ possible worlds

Understanding the Table

The table shows probabilities for combinations like:

- $C \wedge T \wedge X \rightarrow 0.108$
- $C \wedge T \wedge \neg X \rightarrow 0.012$
- $C \wedge \neg T \wedge X \rightarrow 0.072$
(...and so on for all combinations)

👉 Each value = **joint probability of that exact situation**

How to Use It

👉 Example queries:

- $P(\text{Cavity}) \rightarrow$ add all entries where $C = \text{true}$
- $P(\text{Toothache} \mid \text{Cavity}) \rightarrow$ use conditional formula
- $P(\text{XRay} \mid \text{Toothache}) \rightarrow$ sum relevant rows

👉 You just:

Select + add + divide (if needed)

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- **Probability of any proposition computed by finding atomic events where proposition is true and adding their probabilities**
 - $P(\text{cavity} \vee \text{toothache})$
 $= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064$
 $= 0.28$

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- $P(\text{cavity})$
 $= 0.108 + 0.012 + 0.072 + 0.008$
 $= 0.2$

- $P(\text{cavity})$ is called a marginal probability and the process of computing this is called marginalization

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

- Can also compute conditional probabilities.
 - $P(\neg \text{cavity} \mid \text{toothache})$
 $= P(\neg \text{cavity} \wedge \text{toothache}) / P(\text{toothache})$
 $= (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064)$
 $= 0.4$
 - Denominator is viewed as a normalization constant:
 - Stays constant no matter what the value of Cavity is.
 (Book uses α to denote normalization constant $1/P(X)$, for random variable X.)
-

★EXAMPLE:

MACHINE PARTS EXAMPLE – CONDITIONAL PROBABILITY.PDF, EXAMPLE 1

Machine produces parts: 90% good, 2% slightly defective, 8% obviously defective.
Inspection discards obviously defective parts. Quality of shipped parts:

$$P(\text{Good} \mid \text{passes inspection}) = P(\text{Good}) / P(\text{not OD}) = 0.90 / 0.92 \approx 0.978$$



Given Information

- Good parts = 90% → $P(\text{Good}) = 0.90$
- Slightly defective = 2%
- Obviously defective (OD) = 8% → $P(\text{OD}) = 0.08$



Inspection removes **obviously defective parts only**



What Does “Passes Inspection” Mean?

- Pass = NOT obviously defective



So:

- $P(\text{not OD}) = 1 - 0.08 = 0.92$



Formula Used

- $P(\text{Good} \mid \text{passes}) = P(\text{Good} \wedge \text{passes}) / P(\text{passes})$



Since all good parts pass inspection:

- $P(\text{Good} \wedge \text{passes}) = P(\text{Good}) = 0.90$



So:

- $P(\text{Good} \mid \text{passes}) = 0.90 / 0.92 \approx 0.978$



Even though originally 90% were good,
after inspection → **~97.8% of shipped parts are good**

Bayes' Rule & Diagnosis

- Formula:

- $P(a | b) = (P(b | a) \times P(a)) / P(b)$

👉 Meaning:

- Posterior = (Likelihood × Prior) / Normalization

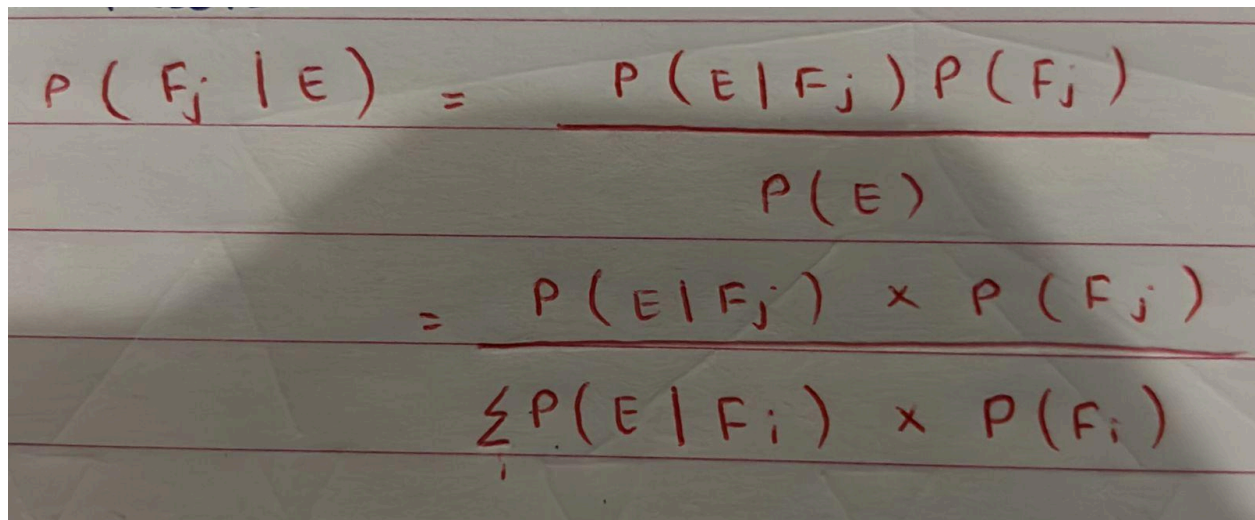
👉 Terms:

- $P(a) \rightarrow$ **Prior** (initial belief)
- $P(b | a) \rightarrow$ **Likelihood** (how likely evidence is if cause is true)
- $P(b) \rightarrow$ **Normalization** (overall probability of evidence)

$$P(A/B) = \frac{P(B/A).P(A)}{P(B)}$$

=> Baye's theorem general form:

$$P(F_i | E) = (P(E | F_i) \times P(F_i)) / \sum_i [P(E | F_i) \times P(F_i)]$$



Handwritten derivation of Bayes' theorem general form:

$$P(F_j | E) = \frac{P(E | F_j) P(F_j)}{P(E)}$$
$$= \frac{P(E | F_j) \times P(F_j)}{\sum_i P(E | F_i) \times P(F_i)}$$

Cause vs Effect (Diagnosis)

- We want:
 - $P(\text{Cause} \mid \text{Effect})$

👉 Using Bayes:

- $P(\text{Cause} \mid \text{Effect}) = (P(\text{Effect} \mid \text{Cause}) \times P(\text{Cause})) / P(\text{Effect})$

👉 Key idea:

We reverse probability → from effect → cause

Example (Disease Diagnosis)

- Disease = TB
- Symptom = coughing

👉 Important points:

- $P(\text{disease} \mid \text{symptom})$ **changes by location** (e.g., USA vs TB-common country)
- $P(\text{symptom} \mid \text{disease})$ is **more stable**

👉 So:

- It's easier to learn:
 - $P(\text{symptom} \mid \text{disease})$

👉 And then use Bayes to compute:

- $P(\text{disease} \mid \text{symptom})$
-

Example (Diagnosis)

- $F_1 = \text{TB}$
- $F_2 = \text{Flu}$
- $E = \text{coughing}$

👉 Then:

- $P(\text{TB} \mid \text{cough}) = \frac{(P(\text{cough} \mid \text{TB}) \times P(\text{TB}))}{[P(\text{cough} \mid \text{TB}) \times P(\text{TB}) + P(\text{cough} \mid \text{Flu}) \times P(\text{Flu})]}$

! What About $P(\text{symptom})$?

- Needed in denominator:
 - $P(\text{disease} \mid \text{symptom}) = \frac{(P(\text{symptom} \mid \text{disease}) \times P(\text{disease}))}{P(\text{symptom})}$

👉 But:

- When comparing diseases, we can **ignore $P(\text{symptom})$**
- Because it's the same for all options

👉 Key idea:

Focus on numerator when comparing probabilities

Conditioning Idea

- Instead of joint probabilities, use conditional ones

👉 Formula:

- **$P(a) = P(a \wedge b) + P(a \wedge \neg b)$**

👉 Replace using conditionals:

- $P(a) = P(a | b) \times P(b) + P(a | \neg b) \times P(\neg b)$
-

General Formula

- $P(Y) = \sum P(Y | z) \times P(z)$

👉 Meaning:

- Sum over all possible causes z

👉 Called:

- **Marginalization + Conditioning**
-

Independence (alr done above)

- Two random variables **A** and **B** are independent if:
 - $P(A | B) = P(A)$
 - $P(B | A) = P(B)$

👉 Meaning:

- Knowing B **does not change** probability of A
-

Equivalent Form

- Independence can also be written as:
 - $P(A \wedge B) = P(A) \times P(B)$

👉 Key idea:

If independent → just multiply probabilities

Why Independence Matters

- Without independence → need full joint distribution (complex)
- With independence → calculations become **simple**

👉 Example:

- Instead of computing $P(A \wedge B)$ from table, just do:
 - $P(A) \times P(B)$

Effect on Probabilistic Inference

- **Reduces complexity**
- **Avoids** storing **huge joint tables**
- Makes **reasoning faster**

👉 Key idea:

Independence = simplification tool in probability

Intuition

- Dependent: events influence each other
- Independent: events behave separately

👉 Example:

- Coin tosses → independent
 - Drawing cards without replacement → dependent
-



Conditional Independence

- **Absolute independence** is **rare in real life**
- Many variables are related (e.g., in dentistry: toothache, X-ray, cavity)

👉 So instead of full independence, we use: **conditional independence**

A and B are conditionally independent given C if:

- $P(A | B, C) = P(A | C)$
- $P(B | A, C) = P(B | C)$

👉 Meaning: gfb

- Once **C** is **known**, **B** gives **no extra info** about **A**



Equivalent Form

- $P(A \wedge B | C) = P(A | C) \times P(B | C)$

👉 Same idea as independence, but **under condition C**

eg A disease affects 2% of population

$$P(D) = 0.02$$

$$\Rightarrow P(\neg D) = 1 - 0.02 = 0.98$$

② A test is performed:

• if the person has the disease, test is positive

$$P(+|D) = 0.95$$

• if the person doesn't have the disease, but is still positive (false).

$$P(+|\neg D) = 0.10$$

Q. a) what is the probability that a normally selected person tests positive:

$$P(+)=?$$

$$P(+)=P(+|D) \cdot P(D) + P(+|\neg D) \cdot P(\neg D)$$

$$\stackrel{\text{Total}}{\text{positive}} = 0.95 \cdot 0.02 + 0.10 \cdot 0.98$$

$$= 0.019 + 0.098$$

$$= 0.117$$

b) if the Person tests positive, what is the probability they actually have the disease?

$$P(D|+) = ?$$

$$P(D|+) = \frac{P(+|D) \cdot P(D)}{P(+)}$$

$$P(+)$$

↓
total

$$= \frac{0.95 \times 0.02}{0.117}$$

$$= 0.1624$$